

Dream-SLAM: Dreaming the Unseen for Active SLAM in Dynamic Environments (Supplementary Material)

Abstract—In this supplementary material, we provide additional analysis and experimental results that are omitted from the main manuscript due to space limitations:

- Handling a broader range of dynamic objects (see Section I).
- Exploration planning in dynamic scenes (see Section II).
- Time costs of full active loop (see Section III).
- Memory usage of the system (see Section IV)
- Sensitivity analysis of the quality of dynamic-object segmentation to our method (see Section V)
- Sensitivity analysis of errors in the underlying Gaussian reconstruction to our method (see Section VI)
- Mapping evaluation in terms of geometry reconstruction (see Section VII)
- Generalization evaluation of the proposed neural network in unseen environments (see Section VIII)
- Reliability analysis of the dreamed content (see Section IX)
- Strategies for incorporating confidence into dreamed Structures and global planning (see Section X)
- Fluctuation evaluation of system performance over multiple runs (see Section XI)

I. HANDLING A BROADER RANGE OF DYNAMIC OBJECTS

In the main manuscript, we focus on non-rigid humans. To demonstrate the generalizability of our approach, in this section, we provide experiments on rigid objects (e.g., boxes), **swinging doors**, and other non-rigid subjects (e.g., animals).

Recall that we predict the attributes of per-pixel Gaussians based on our feedforward network. This network enables the reconstruction of the entire content at a time, which allows us to handle different types of dynamic objects without relying on any shape priors. For validation, we conduct experiments on a sequence of the Bonn dataset [1] that involves a non-rigid human carrying a rigid box. As shown in Fig. 1, our Dream-SLAM can reliably render the human, box, and background from a novel viewpoint across multiple timestamps. In addition, we utilize the Casual-videos dataset [2] to validate the capability in handling non-rigid animals such as dogs and cats. As illustrated in Fig. 2, our Dream-SLAM achieves high-fidelity reconstruction for both dynamic foreground and static background. **Furthermore, we conduct experiment on a self-collected sequence involving a swinging door. The results are shown in Fig. 3. Our method successfully reconstructs the states of both the person and the swinging door at different time steps, while maintaining high-quality novel-view rendering.**

II. EXPLORATION PLANNING IN DYNAMIC SCENES



Fig. 1. Tests of our Dream-SLAM to handle both rigid boxes and non-rigid humans of the Bonn dataset [1]. We present a novel-view rendering result of the reconstructed map on Sequence `ps_track`. The red, green, and blue dotted lines denote the trajectories of the human, camera, and box, respectively.



(a)



(b)

Fig. 2. Tests of our Dream-SLAM to handle non-rigid animals on the Casual-videos dataset [2]. We present novel-view rendering results of the reconstructed maps on (a) Sequence `cat-pikachu-0-0005`, and (b) Sequence `dog-robolounge-0000`. The red and green dotted lines denote the trajectories of the animal and camera, respectively.

In Section VI-B of the main manuscript, we provide explo-

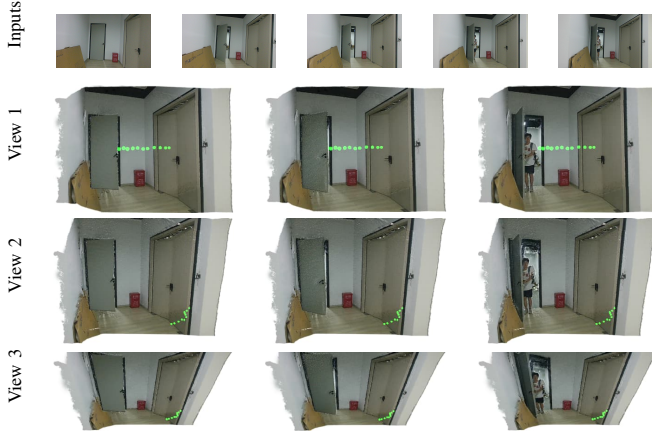


Fig. 3. Application of our method to a swinging door in our self-collected data. The first row shows the sampled input images, and each subsequent row presents the renderings from a novel viewpoint at three different time steps. The green dotted lines denote the camera trajectories.



Fig. 4. A comparison in terms of dynamic scene mapping for planning between ActiveSplat [3] and our Dream-SLAM on Sequence CETmJJqkhcK of the HM3D dataset [4]. (a) ActiveSplat inappropriately reconstructs a dynamic human as static obstacles that occlude the navigable doorway. (b) Our Dream-SLAM reliably reconstructs the dynamic human at different times (the red dotted line denotes the human trajectory). Such a mapping result facilitates effective obstacle avoidance and a thorough scene exploration.

ration planning experiments in dynamic environments. State-of-the-art methods such as ActiveSplat [3] fail to explore the entire scene, while our Dream-SLAM can complete a thorough exploration. We analyze the reasons as follows.

As shown in Fig. 4(a), ActiveSplat fails to distinguish the dynamic foreground from the static background, leading to dynamic humans being reconstructed as static obstacles. The reconstructed humans are not only low-quality, but also block feasible navigation paths (e.g., the doorways). As a result, certain areas in each testing scene remain occluded and unexplored. In contrast, Fig. 4(b) shows that our Dream-SLAM can reconstruct the dynamic human at different times. The robot can effectively avoid the reconstructed human based on the human’s localization and shape information. Moreover, the reconstructed dynamic human does not occlude the path, which is crucial for enabling the thorough exploration of the full scene.

III. TIME COSTS OF FULL ACTIVE LOOP

In the following, we provide a detailed analysis of the computational cost of each module involved in localization and mapping as well as exploration planning, as shown in Fig. 5.

A. Time costs of localization and mapping

The localization and mapping process consists of frontend processing and asynchronous backend optimization. We first analyze the computational cost of the frontend components. We first extract dynamic-object segmentation masks, which takes 0.012 s. The diffusion model then generates the cross-spatio-temporal image in 0.21 s. Although this generation process constitutes the main computational cost, the use of the latent consistency model (LCM) enables efficient inference. Subsequently, the feedforward network predicts the Gaussian representation in 0.068 s. Finally, the predicted Gaussians and the generated cross-spatio-temporal image are jointly used to optimize the camera pose, which requires 0.15 s. These frontend components together result in a total localization and mapping time of 0.44 s per loop.

Similar to ORB-SLAM [5], our map optimization is performed asynchronously in a separate backend thread and therefore does not affect the frontend processing rate. In the backend, we perform multi-view Gaussian refinement for both the background and foreground (see Section IV-C of the main manuscript), which requires 0.42 s on average.

B. Time costs of exploration planning

For exploration planning, semantically plausible image inpainting requires 0.17 s. The feedforward network subsequently predicts the Gaussian representation in 0.068 s. Finally, the planner leverages the dreamed information to determine the exploration path. This stage consists of topological graph construction, subregion clustering, and global path planning, with a total computational cost of 0.05 s. Overall, these components result in an exploration planning time of approximately 0.29 s per loop.

IV. MEMORY USAGE OF THE SYSTEM

The GPU memory consumption of our system mainly comes from neural network inference and the optimization of Gaussian representations and camera poses. Since different modules are activated at different stages of the active loop, the overall memory usage varies slightly during operation. In our experiments, the total GPU memory consumption ranges from approximately 10.3 GB to 10.6 GB, which is on the same level as existing Gaussian-based SLAM methods [6], [7]. A component-wise breakdown of the memory usage is reported in Table I.

For the feedforward Gaussian prediction network, the additional Gaussian head introduces extra memory consumption compared with DUST3R [8], while the overall memory overhead remains reasonable. For the diffusion modules, both are fine-tuned from Stable Diffusion v2 [9] using LoRA. Therefore, we keep only one Stable Diffusion v2 backbone

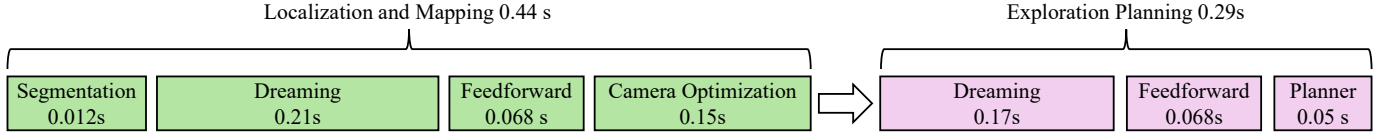


Fig. 5. Average runtime of our Dream-SLAM over the full active loop. The computational cost is divided into localization and mapping as well as exploration planning, with a detailed breakdown of the components in each stage.

TABLE I
GPU MEMORY COSTS OF DIFFERENT MODULES OF OUR METHOD.

Type	GPU memory cost (MB)
Our Gaussian prediction network	4155.30
Mask R-CNN [10] / YOLOE [11]	3.22 / 104.89
Diffusion model	4824.21
LoRA for cross-spatio-temporal images	12.45
LoRA for semantically plausible image	12.45
Our Gaussian and camera pose-related optimization	1442.56 ~ 1736.23

TABLE II
WE APPLY DILATION AND EROSION AT RATIOS OF 10%, 30%, AND 50% TO THE FOREGROUND SEGMENTATION MASKS AND EVALUATE THEIR EFFECTS ON CAMERA LOCALIZATION ON THE `ps_track` SEQUENCE OF THE BONN DATASET [1] AND THE `wk_xyz` SEQUENCE OF THE TUM DATASET [12].

	Seq. <code>ps_track</code>		Seq. <code>wk_xyz</code>	
	RMSE	SD	RMSE	SD
Original	1.51	0.55	1.74	0.75
Dilation 10%	1.51	0.55	1.74	0.76
Dilation 30%	1.51	0.56	1.78	0.78
Dilation 50%	1.52	0.58	1.80	0.79
Erosion 10%	1.51	0.55	1.74	0.75
Erosion 30%	1.52	0.55	1.76	0.75
Erosion 50%	1.54	0.56	1.77	0.78

in GPU memory and dynamically switch between the two LoRA weights for cross-spatio-temporal image and semantically plausible structure generation. This avoids simultaneously loading two complete diffusion models and substantially reduces GPU memory consumption. For foreground segmentation, we employ either Mask R-CNN [10] or YOLOE [11], both of which introduce relatively low additional GPU memory overhead. Unlike the nearly constant memory consumption of these neural network modules, the memory required for Gaussian and camera-parameter optimization varies during operation. This variation mainly arises because the number of Gaussian primitives and optimization variables changes as new frames are incorporated into the map. To control the growth of the map and its associated memory consumption, our backend periodically prunes redundant Gaussian primitives during map optimization. Consequently, the overall GPU memory usage remains within a relatively stable range.

TABLE III
ABLATION STUDY OF OUR OPTIMIZATION ON THE BONN DATASET [1]. WE REPORT THE MESH ALIGNMENT ERROR IN CENTIMETERS, FOLLOWING THE EVALUATION PROTOCOL OF [13], [14].

Sequences of Bonn Dataset	With Optimization	Without Optimization
<code>ps_track</code>	8.15	10.08
<code>ps_track2</code>	6.07	11.21
<code>balloon</code>	5.56	6.63
<code>balloon2</code>	8.36	11.22
<code>mv_box</code>	6.47	6.81
<code>mv_box2</code>	7.26	8.37
Average	6.98	9.05

V. SENSITIVITY ANALYSIS OF THE QUALITY OF DYNAMIC-OBJECT SEGMENTATION TO OUR METHOD

To analyze this sensitivity, we conduct ablation experiments on the `ps_track` sequence of the Bonn dataset [1] and the `wk_xyz` sequence of the TUM dataset [12]. Specifically, we perturb the foreground masks by applying dilation and erosion at ratios of 10%, 30%, and 50% and evaluate the resulting changes in localization accuracy. As shown in Table II, a 10% mask perturbation causes no noticeable change in localization accuracy, while a 30% perturbation results in only a slight degradation. The accuracy decreases more substantially when the perturbation ratio reaches 50%. Since the 30% and 50% perturbations represent relatively severe segmentation errors, these results indicate that our method is robust to the moderate mask inaccuracies typically encountered in practical segmentation.

As for inter-frame mask tracking, similar to PG-SLAM [7], we use optical flow to establish the correspondence of dynamic-object masks across frames.

VI. SENSITIVITY ANALYSIS OF ERRORS IN THE UNDERLYING GAUSSIAN RECONSTRUCTION TO OUR METHOD

The underlying Gaussian reconstruction is produced by a feedforward network followed by backend optimization. To evaluate the quality of the feedforward prediction and the effectiveness of backend optimization, we compare the reconstructed meshes obtained before and after optimization against the ground-truth meshes provided by the Bonn dataset [1]. Following the evaluation protocol in [14], [15], we adopt the mesh alignment error as the reconstruction metric. Specifically, for each vertex of the reconstructed mesh, we compute its minimum Euclidean distance to the ground-truth surface and

TABLE IV
MESH ACCURACY COMPARISON OF OUR DREAM-SLAM WITH PG-SLAM [7] AND RoDyn-SLAM [14] ON THE BONN DATASET [1]. WE REPORT THE MESH ALIGNMENT ERROR IN CENTIMETERS, FOLLOWING THE EVALUATION PROTOCOL OF [13], [14].

Sequences of Bonn Dataset	Dream-SLAM (our)	PG-SLAM [7]	RoDyn-SLAM [14]
ps_track	8.15	8.40	11.32
ps_track2	6.07	7.26	12.58
balloon	5.56	8.49	9.87
balloon2	8.36	9.22	14.03
mv_box	6.47	6.83	10.68
mv_box2	7.26	7.92	10.68

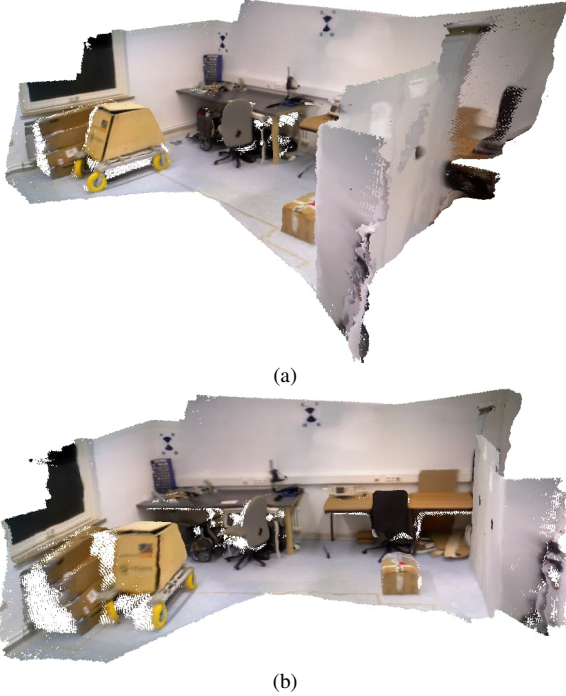


Fig. 6. Mesh reconstructed by our Dream-SLAM on Sequence mv_box2 of Bonn dataset [1]. We provide observations from two random viewpoints.

report the mean distance over all vertices. As reported in Table III, the feedforward estimation alone achieves an average mesh alignment error of 9.05 cm. Reconstruction accuracy at this scale has already been shown to provide sufficiently reliable scene representations, as demonstrated by [7], [14]. The subsequent backend optimization further reduces the reconstruction error and improves the geometric accuracy and reliability of the Gaussian map.

VII. MAPPING EVALUATION IN TERMS OF GEOMETRY RECONSTRUCTION

We assess the quality of the Gaussian reconstruction both qualitatively and quantitatively. For the qualitative evaluation, we visualize the reconstructed maps from two different viewpoints. As shown in Fig. 6, our method accurately reconstructs both the scene geometry and appearance. We further perform a quantitative comparison against the ground-truth meshes provided by the Bonn dataset [1]. As reported in Table IV, our Dream-SLAM consistently achieves lower mesh alignment

TABLE V
QUANTITATIVE RENDERING PERFORMANCE (PSNR, SSIM, AND LPIPS) OF CROSS-SPATIO-TEMPORAL IMAGES ON EMDB [16] AND DYCHECK [17] DATASETS.

Dataset	Sequence	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Emdb [16]	13_outdoor_long_walk	24.42	0.873	0.241
	74_outdoor_rom	23.94	0.906	0.151
Dycheck [17]	haru	21.95	0.889	0.140
	sriracha-tree	23.95	0.923	0.124
Average		23.57	0.898	0.164

errors than PG-SLAM [7] and RoDyn-SLAM [14] across all evaluated sequences.

VIII. GENERALIZATION EVALUATION OF THE PROPOSED NEURAL NETWORK IN UNSEEN ENVIRONMENTS

The overall generalization capability of our method relies on the generalization of three key modules, i.e., cross-spatio-temporal image generation, semantically plausible structure generation, and feedforward reconstruction. In the following, we provide detailed analyses of these three modules.

A. Cross-spatio-temporal Images

Recall that we train the cross-spatio-temporal image generation model on the WildGS [6] and Neuman [18] datasets (see Section VI-A of the main manuscript). We evaluate its generalization capability on the EMDB [16] and DyCheck [17] datasets. These datasets cover both indoor and outdoor scenes, as well as different dynamic-object categories and motion patterns, including humans, cats, and dogs. For quantitative evaluation, we construct the corresponding ground-truth images following the procedure introduced in the Section IV-A of the main manuscript, and then compute image-quality metrics. As reported in Table V, our method achieves an average PSNR of 23.57 dB across different sequences of the two datasets, indicating that the generated content maintains sufficient quality for subsequent localization. We further sample two representative examples from each sequence for qualitative evaluation, as shown in Fig. 7. The generated foreground content exhibits strong visual consistency with the corresponding ground truth.

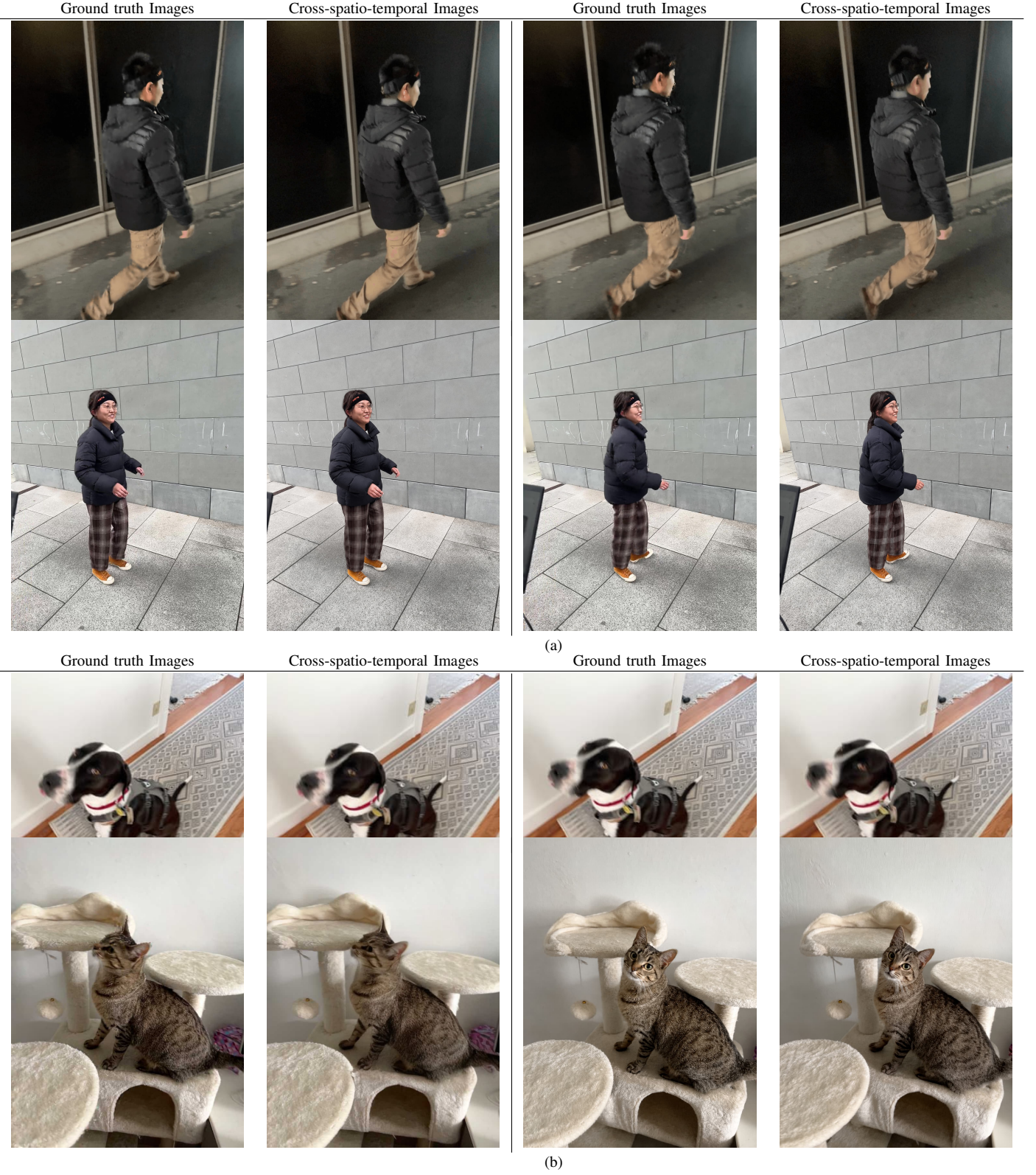


Fig. 7. Qualitative generalization evaluation of cross-spatio-temporal image generation. Generated images are qualitatively compared with the corresponding ground-truth images on (a) the 13_outdoor_long_walk and 74_outdoor_rom sequences of the EMDB dataset [16], and (b) the haru and sriracha-tree sequences of the DyCheck dataset [17], achieving high visual consistency.

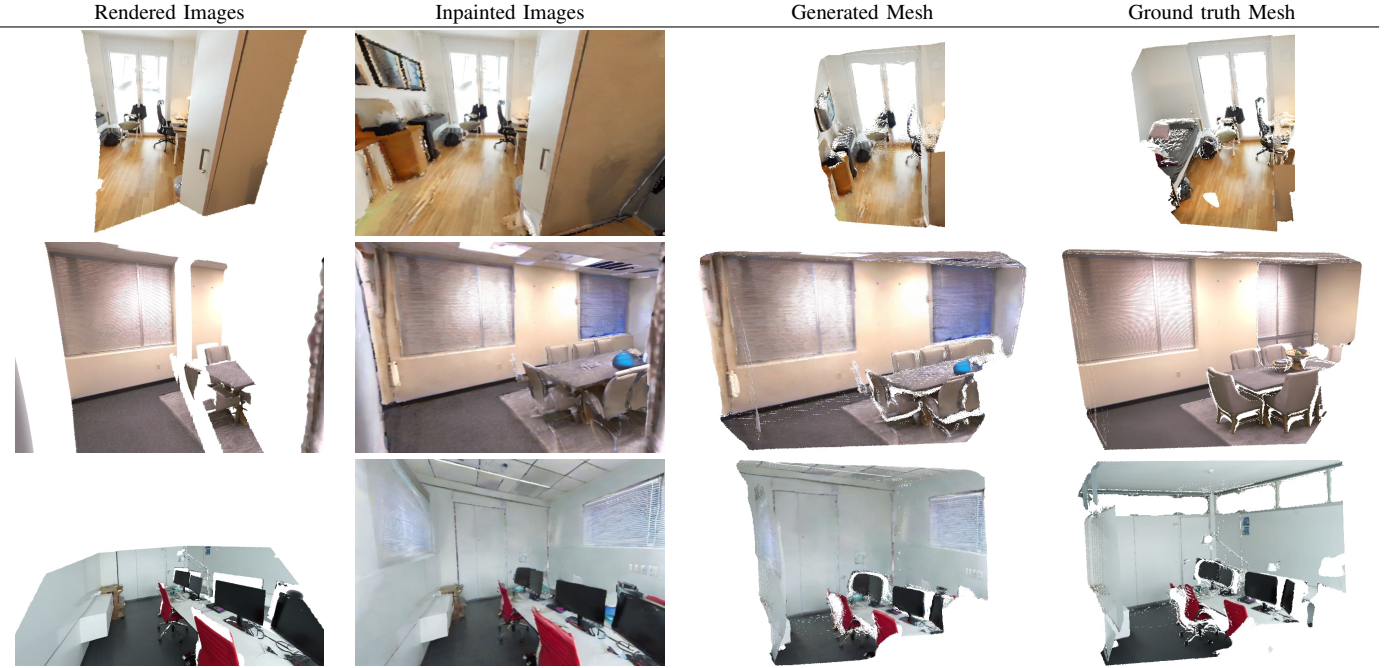


Fig. 8. Qualitative generalization evaluation of semantically plausible structures. We present the rendered images, the inpainted images, and the resulting generated 3D structures, and compare the generated structures with the corresponding ground-truth structures on the 0d2ee665be and e0abd740ba sequences of the ScanNet++ dataset [19] and the apart1 sequence of the Replica dataset [20]. Although the generated mesh textures are not fully consistent with the ground truth, the overall structural layouts remain similar, which is sufficient for effective downstream planning.

B. Semantically Plausible Structures

Recall that we train the semantically plausible structures model on the Matterport3D [21] dataset (see Section VI-B of the main). We further evaluate semantically plausible structure generation on the ScanNet++ [19] and Replica [20] datasets, which were excluded from training and exhibit scene layouts different from those in the training data. Since our goal is not to reconstruct structures that exactly match the ground truth, but rather to generate semantically plausible structures that can support exploration planning, direct quantitative comparison is less suitable for this evaluation. Instead, we provide qualitative comparisons among the rendered images (white pixels correspond to unexplored areas), the inpainted images, the generated 3D structures, and the corresponding ground-truth structures, as shown in Fig. 8. Specifically, our network first predicts plausible image content conditioned on the contextual information in the rendered view, and the feedforward network then converts the generated image into a 3D structure. Although the generated structures may not exactly match the ground truth in texture, their overall spatial layouts remain similar. This generalization capability is sufficient to provide reliable structural guidance for downstream planning.

C. Feedforward Gaussian Prediction Network

As mentioned in Section VI-A of the main manuscript, for the training data, we use the datasets adopted by MonST3R [22] and additionally include the NeuMan [18] and WildGS [6] datasets. We further evaluate the generalization capability of the feedforward Gaussian prediction network on the EMDB [16] and DyCheck [17] datasets. The qualitative

TABLE VI
QUANTITATIVE RENDERING PERFORMANCE (PSNR, SSIM, AND LPIPS)
OF OUR PROPOSED FEEDFORWARD NETWORK ON EMDB [16] AND
DYCHECK [17] DATASETS.

Dataset	Sequence	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Emdb [16]	13_outdoor_long_walk	24.52	0.896	0.117
	74_outdoor_rom	24.50	0.928	0.119
Dycheck [17]	haru	26.23	0.947	0.096
	sriracha-tree	26.69	0.923	0.124
Average		25.49	0.924	0.114

results are shown in Fig. 9, where we visualize the reconstruction under the input view together with the rendered results from two novel views. These examples cover both indoor and outdoor scenes, as well as different dynamic-object categories such as humans and animals. In all cases, our method reconstructs high-quality Gaussian representations and achieves favorable novel-view rendering quality. For quantitative evaluation, we report PSNR, SSIM, and LPIPS in Table VI. The strong quantitative results further demonstrate the generalization capability of our method beyond the training distribution.

IX. RELIABILITY ANALYSIS OF THE DREAMED CONTENT

In the main manuscript, we have partly demonstrated the high reliability of our proposed dreaming strategy, since the dreamed content can improve localization and mapping accuracy as well as exploration efficiency. In this section, we further investigate the reliability of the dreamed content, i.e., cross-spatio-temporal images and semantically plausible structures.

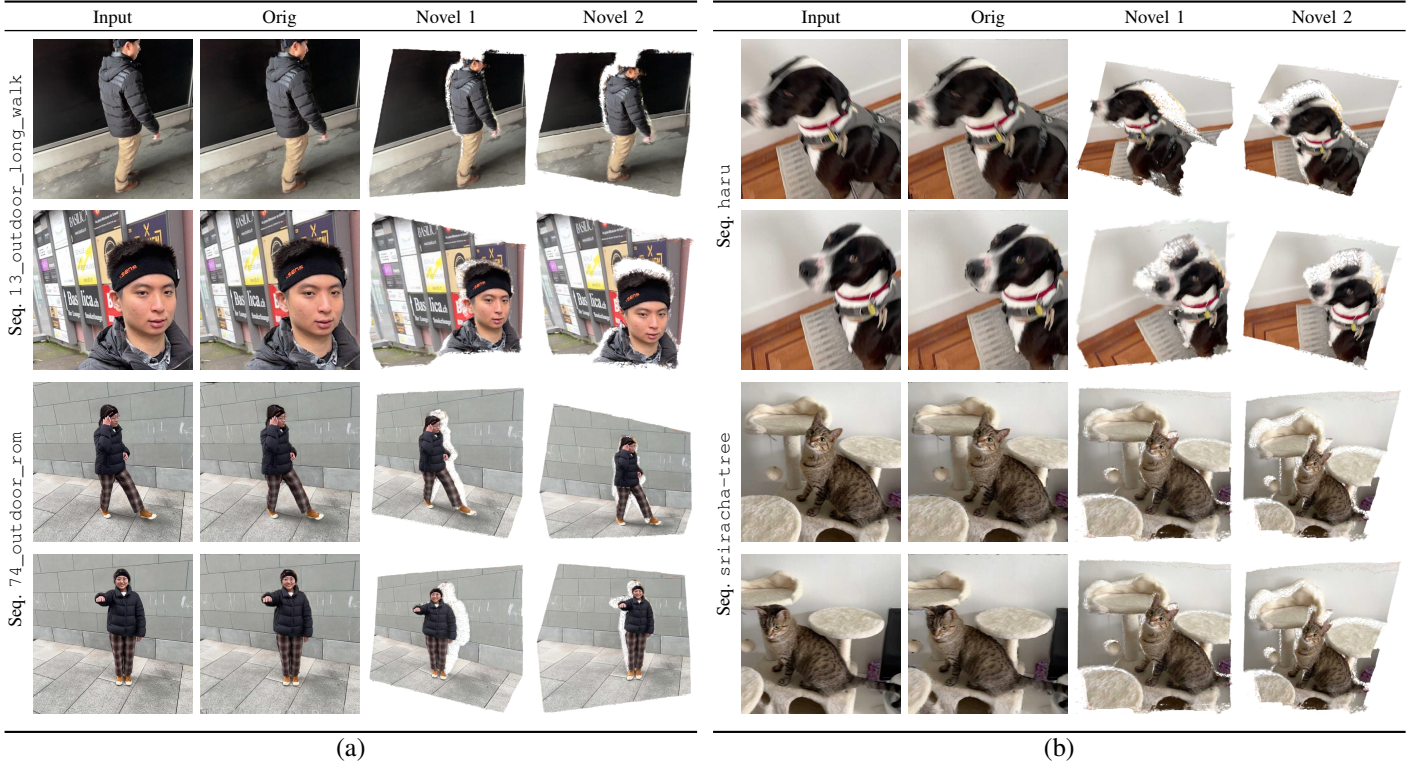


Fig. 9. Qualitative generalization evaluation of the feedforward network. Rendering results are reported on (a) the 13_outdoor_long_walk and 74_outdoor_rom sequences of the EMDB dataset [16], and (b) the haru and sriracha-tree sequences of the DyCheck dataset [17]. From left to right, each row shows the input image, the rendering from the original viewpoint, and the renderings from two novel viewpoints.

TABLE VII

PLANNING PERFORMANCE COMPARISON BETWEEN THE ORIGINAL OCCUPANCY-SCORE-BASED FILTERING STRATEGY AND THE CONFIDENCE-AWARE PLANNING VARIANT.

Sequence	Swormville of Gibson [23]		4h4JxvG3cip of HM3D [4]	
	CR↑ (%)	PL↓ (m)	CR↑ (%)	PL↓ (m)
Occupancy Filtering	95.9	51.6	95.1	89.6
Confidence-aware Planning	95.9	55.2	95.6	83.7

A. Cross-spatio-temporal Images

In general, our cross-spatio-temporal images exhibit high reliability. The direct and indirect evaluations are presented in Section VIII above and Section VI of the manuscript, respectively.

Despite this overall reliability, very occasional low-quality generations may still occur. As shown in Fig. 10(a), the neural network may sometimes hallucinate incorrect foreground textures, leading to partially inaccurate foreground appearance. To avoid the effects of outliers during SLAM, we introduce a self-adaptive and fast outlier identification strategy based on optical flow. We compute the optical-flow tracking consistency between the real observation I_t and the cross-spatio-temporal image I_{t+1}^t , as these two images contain the foreground object in the same state. Fig. 10(c) shows that erroneous generated content leads to inconsistent tracking in some foreground

regions, resulting in unmatched areas highlighted in red. In contrast, as shown in Fig. 10(f), when the dreamed content is reliable, the foreground regions achieve consistent optical tracking, with most areas correctly matched and highlighted in green. Fig. 10(g) systematically demonstrates the effectiveness of our outlier identification strategy on the ps track sequence of the Bonn dataset [1]. The matching ratio of most image pairs is higher than 90% and close to 100%, with only one observed hallucination case at approximately 60% (see the magenta circle). Since this hallucination case exhibits a clear deviation from the normal matching ratio distribution, it can be easily identified and filtered out.

B. Semantically Plausible Structures

Our objective is not to reconstruct unexplored regions that exactly match the ground truth. Instead, we aim to dream semantically plausible structures that provide useful spatial cues for exploration planning and can reliably support subsequent planning decisions. To this end, we conduct extensive validation and provide two representative cases on the 4h4JxvG3cip and 6HMiY15cxis sequences of the HM3D dataset [4], as shown in Fig. 11. For each sequence, we visualize the map used for planning, the next selected target, and the resulting global trajectory. Although the dreamed structures do not exactly match the ground truth, they preserve semantically plausible layouts and the spatial information required for planning. By predicting possible structures in unexplored regions, the dreaming strategy provides additional guidance for selecting targets with greater global exploration

TABLE VIII

STATISTICAL COMPARISON OF OUR DREAM-SLAM, MAPEx [24], AND ACTIVESPLAT [3]. WE RUN EACH METHOD FIVE TIMES ON THE T7nCRMUFNR SEQUENCE OF THE HM3D DATASET [4] AND THE SWORMVILLE SEQUENCE OF THE GIBSON DATASET [23], RESPECTIVELY. FOR EACH SEQUENCE, CR (%) AND PL (M) ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION OVER FIVE RUNS, TOGETHER WITH THE p -VALUES RELATIVE TO OUR METHOD.

Sequences	Metric	Method	Individual Runs					Mean $\pm$ Std	p -value
			#1	#2	#3	#4	#5		
T7nCRMUFNR	CR $\uparrow$	MapEx [24]	95.2	95.7	94.9	95.0	95.0	95.16 $\pm$ 0.32	<0.001
		ActiveSplat [3]	94.3	94.9	94.6	94.7	94.2	94.54 $\pm$ 0.29	<0.001
		Dream-SLAM (ours)	96.8	96.8	96.9	96.9	96.7	96.82 $\pm$ 0.08	-
	PL $\downarrow$	MapEx [24]	98.0	102.7	104.6	101.3	102.6	101.84 $\pm$ 2.45	<0.001
		ActiveSplat [3]	109.3	108.0	103.7	104.2	108.3	106.70 $\pm$ 2.56	<0.001
		Dream-SLAM (ours)	75.9	73.7	73.9	78.1	76.2	75.56 $\pm$ 1.82	-
Swormville	CR $\uparrow$	MapEx [24]	95.6	94.6	95.4	95.7	95.2	95.30 $\pm$ 0.44	0.013
		ActiveSplat [3]	95.3	95.3	94.0	94.7	95.3	94.92 $\pm$ 0.58	0.005
		Dream-SLAM (ours)	95.9	96.2	95.8	96.0	95.9	95.96 $\pm$ 0.15	-
	PL $\downarrow$	MapEx [24]	79.4	79.1	82.8	78.3	81.0	80.12 $\pm$ 1.79	<0.001
		ActiveSplat [3]	91.9	90.2	92.7	92.7	89.4	91.38 $\pm$ 1.51	<0.001
		Dream-SLAM (ours)	51.6	49.3	53.1	50.5	50.2	50.94 $\pm$ 1.46	-

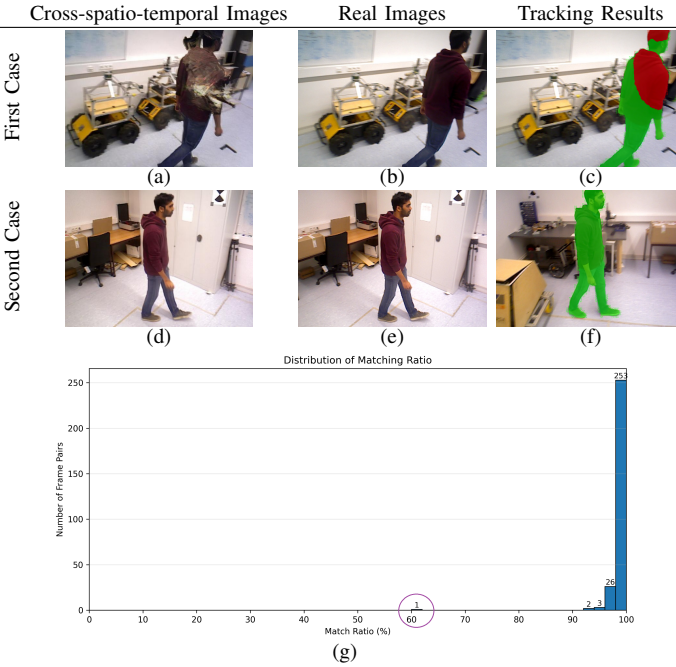


Fig. 10. Comparison analysis of failure and success cases in cross-spatio-temporal image generation. The first row demonstrates a failure case, while the second row illustrates a successful case. (a) and (d) show the generated cross-spatio-temporal images I_{t+1}^t . (b) and (e) show the real observation image I_t . (c) and (f) present the foreground optical tracking results between I_t and I_{t+1}^t , which are used to evaluate the reliability of I_{t+1}^t ; green and red denote matched and unmatched regions, respectively. (g) reports the statistics of the match ratio on the ps_track sequence of the Bonn dataset [1], showing that such hallucination cases occur only occasionally and can be effectively filtered out.

gain, leading to more efficient exploration trajectories. These cases demonstrate that the dreamed structures are sufficiently reliable for supporting long-horizon planning.

X. STRATEGIES FOR INCORPORATING CONFIDENCE INTO DREAMED STRUCTURES AND GLOBAL PLANNING

In this section, we first explain how we estimate the confidence of dreamed structures in our original method. We

then introduce a strategy to further incorporate this confidence information into global planning.

A. Confidence of Dreamed Structures

In our framework, the occupancy score serves as a confidence-related indicator for assessing whether a candidate view can support reliable dreaming, as introduced in Section V-B of the main manuscript. A low occupancy score indicates insufficient observed context, resulting in low confidence in the dreamed structure. As the occupancy score increases, more contextual information becomes available to constrain the generation, thereby improving its reliability. Meanwhile, when the occupancy score becomes too high, only a small unexplored region remains and additional dreaming provides limited benefit. Therefore, we select views within an appropriate occupancy-score range for dreaming. In this sense, the occupancy score provides a confidence-aware mechanism that prevents unreliable dreamed structures from being incorporated into subsequent global planning.

We further consider uncertainty as confidence score based on uncertainty estimation methods for diffusion models, such as BayesDiff [25]. These methods introduce probabilistic modeling into the diffusion network to estimate predictive uncertainty. For the same input, multiple stochastic predictions are generated. The variance across these predictions is then used as the uncertainty estimate. However, this process requires repeated sampling for each input and therefore introduces additional computational cost. This is not well suited to our active SLAM setting, where dreaming needs to be performed online. In contrast, our occupancy score provides a lightweight confidence proxy without requiring repeated diffusion inference. The experimental results also show that this mechanism is sufficient to select reliable dreamed structures for subsequent planning.

B. Confidence-aware Global Planning

In our original work, we do not incorporate confidence-aware information gain into global planning. Instead, before

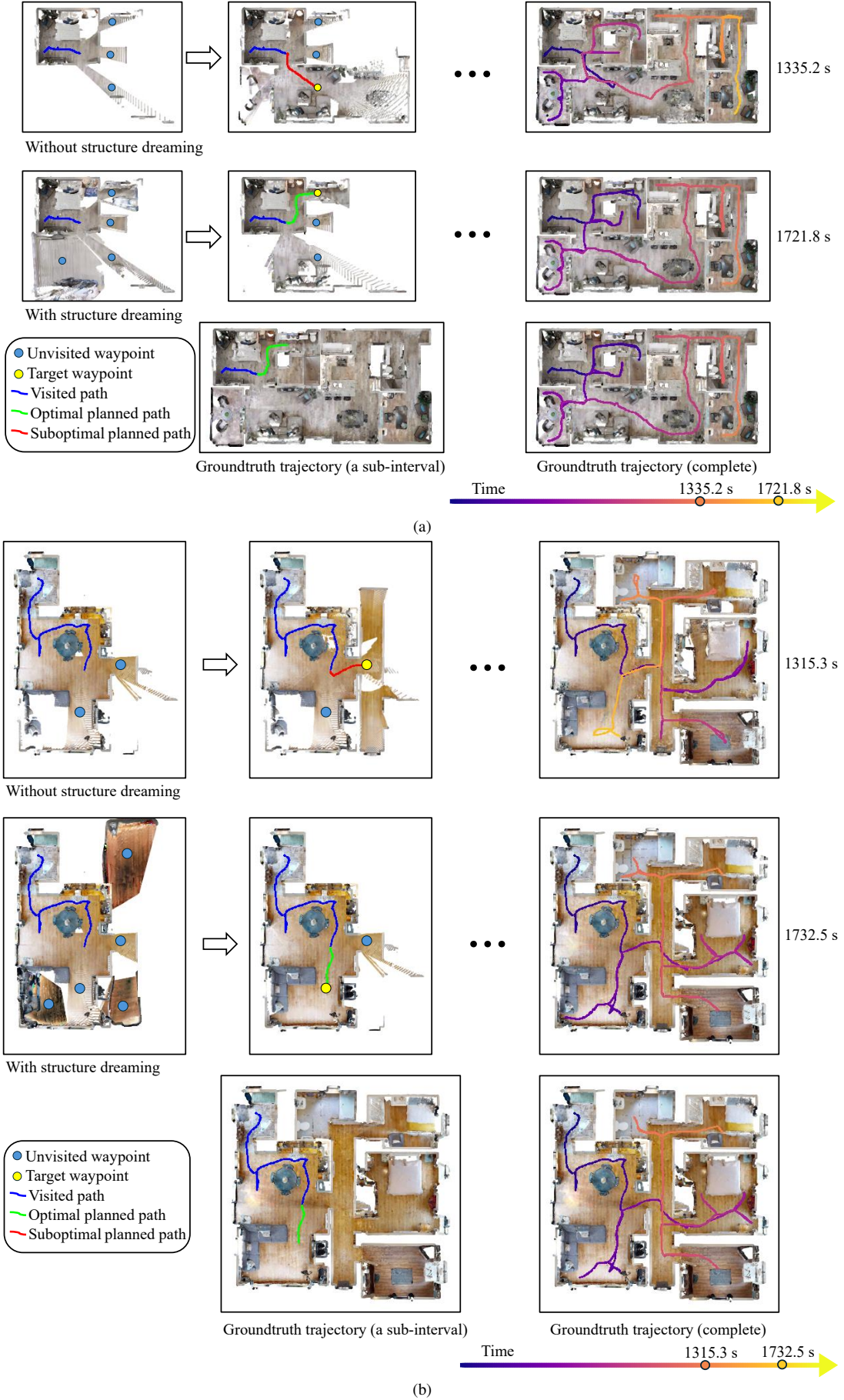


Fig. 11. Effectiveness of dreamed structures for exploration planning. Exploration trajectories with and without the dreaming strategy are compared on two sequences of the HM3D dataset [4]: (a) 4h4JxvG3cip and (b) 6HMiy15cxis.

global planning, we filter unreliable dreamed structures using the aforementioned occupancy score. The remaining structures are then treated equally during global planning based on classical TSP. In other words, all retained structures are assigned a confidence of 1.

We further implement a confidence-aware planning variant. Following [26], we formulate global planning as a Probabilistic Traveling Salesman Problem (PTSP). For each dreamed node, its existence probability is determined by the corresponding confidence score (computed by the above Bayes-Diff). A high-confidence node is more likely to correspond to a valid unexplored region. In contrast, a low-confidence node is considered less reliable and may not actually exist. We compare the original filtering strategy with this confidence-aware variant on the 4h4JxvG3cip sequence of the HM3D dataset [4] and the Swormville sequence of the Gibson dataset [23]. The results are reported in Table VII. On Swormville, the confidence-aware variant achieves slightly better performance than our original filtering strategy. This is because a fixed filtering criterion cannot always be optimal across different scenes. Therefore, on some sequences, directly discarding structures according to a predefined threshold may lead to slightly inferior performance compared with confidence-aware planning. On T7nCRmufFNR, the original filtering strategy performs better than the confidence-aware variant. This is likely because unreliable structures are fully removed and therefore cannot influence the global planning process. In contrast, the confidence-aware formulation still retains their effects through nonzero existence probabilities.

XI. FLUCTUATION EVALUATION OF SYSTEM PERFORMANCE OVER MULTIPLE RUNS

We conduct experiments on the T7nCRmufFNR sequence of the HM3D dataset [4] and the Swormville sequence of the Gibson dataset [23]. For each sequence, we independently run Dream-SLAM, MapEx [24], and ActiveSplat [3] five times. Table VIII reports the CR and PL results of all five runs, together with their mean and standard deviation. We further perform paired t-tests between our method and each baseline and report the corresponding p -values. All reported p -values are below 0.05, indicating that the improvements achieved by Dream-SLAM are statistically significant.

REFERENCES

- [1] E. Palazzolo, J. Behley, P. Lottes, P. Giguère, and C. Stachniss, “ReFusion: 3D Reconstruction in Dynamic Environments for RGB-D Cameras Exploiting Residuals,” in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2019, pp. 7855–7862.
- [2] G. Yang, M. Vo, N. Neverova, D. Ramanan, A. Vedaldi, and H. Joo, “Banmo: Building animatable 3d neural models from many casual videos,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 2863–2873.
- [3] Y. Li, Z. Kuang, T. Li, Q. Hao, Z. Yan, G. Zhou, and S. Zhang, “ActiveSplat: High-Fidelity Scene Reconstruction Through Active Gaussian Splatting,” *IEEE Robotics and Automation Letters*, vol. 10, no. 8, pp. 8099–8106, 2025.
- [4] S. K. Ramakrishnan, A. Gokaslan, E. Wijmans, O. Maksymets, A. Clegg, J. M. Turner, E. Undersander, W. Galuba, A. Westbury, A. X. Chang, M. Savva, Y. Zhao, and D. Batra, “Habitat-Matterport 3D Dataset (HM3D): 1000 Large-scale 3D Environments for Embodied AI,” in *Proceedings of the conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2021.
- [5] C. Campos, R. Elvira, J. J. G. Rodríguez, J. M. M. Montiel, and J. D. Tardós, “ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial, and Multimap SLAM,” *IEEE Transactions on Robotics*, vol. 37, no. 6, pp. 1874–1890, Dec. 2021.
- [6] J. Zheng, Z. Zhu, V. Bieri, M. Pollefeys, S. Peng, and I. Armeni, “WildGS-SLAM: Monocular Gaussian Splatting SLAM in Dynamic Environments,” in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 11 461–11 471.
- [7] H. Li, X. Meng, X. Zuo, Z. Liu, H. Wang, and D. Cremers, “PG-SLAM: Photorealistic and Geometry-Aware RGB-D SLAM in Dynamic Environments,” *IEEE Transactions on Robotics*, vol. 41, pp. 6084–6101, 2025.
- [8] S. Wang, V. Leroy, Y. Cabon, B. Chidlovskii, and J. Revaud, “DUST3R: Geometric 3D Vision Made Easy,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 20 697–20 709.
- [9] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, “High-resolution image synthesis with latent diffusion models,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 10 684–10 695.
- [10] K. He, G. Gkioxari, P. Dollár, and R. Girshick, “Mask R-CNN,” in *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 2980–2988.
- [11] A. Wang, L. Liu, H. Chen, Z. Lin, J. Han, and G. Ding, “Yoloe: Real-time seeing anythi,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 24 591–24 602.
- [12] J. Sturm, N. Engelhard, F. Endres, W. Burgard, and D. Cremers, “A benchmark for the evaluation of RGB-D SLAM systems,” in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2012, pp. 573–580.
- [13] Z. Zhu, S. Peng, V. Larsson, W. Xu, H. Bao, Z. Cui, M. R. Oswald, and M. Pollefeys, “NICE-SLAM: Neural Implicit Scalable Encoding for SLAM,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2022, pp. 12 786–12 796.
- [14] H. Jiang, Y. Xu, K. Li, J. Feng, and L. Zhang, “RoDyn-SLAM: Robust Dynamic Dense RGB-D SLAM with Neural Radiance Fields,” *IEEE Robotics and Automation Letters*, vol. 9, no. 9, pp. 7509–7516, Sept. 2024.
- [15] M. M. Johari, C. Carta, and F. Fleuret, “ESLAM: Efficient Dense SLAM System Based on Hybrid Representation of Signed Distance Fields,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 17 408–17 419.
- [16] M. Kaufmann, J. Song, C. Guo, K. Shen, T. Jiang, C. Tang, J. J. Zárata, and O. Hilliges, “Emdb: The electromagnetic database of global 3d human pose and shape in the wild,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 14 586–14 597.
- [17] H. Gao, R. Li, S. Tulsiani, B. Russell, and A. Kanazawa, “Monocular dynamic view synthesis: A reality check,” *Advances in Neural Information Processing Systems*, pp. 33 768–33 780, Nov. 2022.
- [18] W. Jiang, K. M. Yi, G. Samei, O. Tuzel, and A. Ranjan, “NeuMan: Neural Human Radiance Field from a Single Video,” in *Proceedings of the European Conference on Computer Vision*, 2022, pp. 402–418.
- [19] C. Yeshwanth, Y.-C. Liu, M. Nießner, and A. Dai, “Scannet++: A high-fidelity dataset of 3d indoor scenes,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 12–22.
- [20] J. Straub, T. Whelan, L. Ma, Y. Chen, E. Wijmans, S. Green, J. J. Engel, R. Mur-Artal, C. Ren, S. Verma *et al.*, “The replica dataset: A digital replica of indoor spaces,” *arXiv preprint arXiv:1906.05797*, 2019.
- [21] A. Chang, A. Dai, T. Funkhouser, M. Halber, M. Niebner, M. Savva, S. Song, A. Zeng, and Y. Zhang, “Matterport3D: Learning from RGB-D Data in Indoor Environments,” in *Proceedings of the International Conference on 3D Vision*, 2017, pp. 667–676.
- [22] J. Zhang, C. Herrmann, J. Hur, V. Jampani, T. Darrell, F. Cole, D. Sun, and M.-H. Yang, “MonST3R: A Simple Approach for Estimating Geometry in the Presence of Motion,” in *Proceedings of the International Conference on Learning Representations*, 2025.
- [23] F. Xia, A. R. Zamir, Z. He, A. Sax, J. Malik, and S. Savarese, “Gibson Env: Real-World Perception for Embodied Agents,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2018, pp. 9068–9079.

- [24] C. Ho, S. Kim, B. Moon, A. Parandekar, N. Harutyunyan, C. Wang, K. Sycara, G. Best, and S. Scherer, “MapEx: Indoor Structure Exploration with Probabilistic Information Gain from Global Map Predictions,” in *Proceedings of the IEEE International Conference on Robotics and Automation*, 2025, pp. 13 074–13 080.
- [25] S. Kou, L. Gan, D. Wang, C. Li, and Z. Deng, “Bayesdiff: Estimating pixel-wise uncertainty in diffusion via bayesian inference,” in *Proceedings of the International Conference on Learning Representations*, 2023.
- [26] Y. Lyu, C. Cao, J. Zhang, H. Choset, and Z. Ren, “Rpt*: Global planning with probabilistic terminals for target search in complex environments,” *arXiv preprint arXiv:2601.12701*, 2026.